UNIVERSITY OF CAMBRIDGE

GenAI

Focus of the day: GenAI

- Definition of GenAI
- Prompt-engineering
- ChatGPT Exercises
- Classroom discussion on economic impact of GenAI

PROMPT-ENGINEERING

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- **Prompt**= set of instructions provided to a LLM that programs the LLM by customizing/enhancing or refining its capabilities**; it is much more than asking a question!!!!**
- **Prompt engineering**= how LLMs are programmed via prompts
- **Prompt patterns** are a knowledge transfer method; effective approaches to program LLMs for particular outcomes; “reusable solutions to a recurring problems/tasks”
- Known challenges: training period/sample and impact on answer; size limitations (increasingly less problematic); hallucination

PROMPT-PATTERNS

PATTERN	CONTEXTUAL STATEMENTS
Meta Language Creation Pattern	When I say X, I mean Y
The Flipped Interaction Pattern	I would like you to ask me questions to achieve X
The Persona Pattern	Act as persona X; Provide outputs that persona X would create
The Question Refinement Pattern	Within scope X, suggest a better version of the question to use instead
The Cognitive Verifier Pattern	When you are asked questions, follow the rules; generate a number of additional questions that would help more accurately answer the questions; combine the answers to the individual questions to produce the final answer to the overall question
The Fact Check List Pattern	Generate a set of facts that are contained in the output

PROMPT-PATTERNS

PATTERN	CONTEXTUAL STATEMENTS
The Refusal Breaker Pattern	Whenever you cannot answer the question, explain why you can't answer the question
The Context Manager Pattern	Within scope X, please consider Y and please ignore Y
The Reflection Pattern	Whenever you generate the answer, explain the answer and assumptions behind the answer

Source: *A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT*, Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert,Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt

CLASSROOM EXERCISE

GENAI-PROMPT ENGINEERING

Using the paper Measuring News Sentiment we covered in Lecture 2, use prompt engineering to perform the following tasks:

- Summarise its content in less than 500 tokens, then less than 400 words.
- Provide a critical evaluation of the paper for practical use in economics
- Suggest ways to enhance the models and their accuracy

CLASSROOM DISCUSSION: GENAI and ECONOMIC IMPACT

Pre-Read

- **GPTs are GPTs: An early look at the labor market impact potential of large language models by Tyna Eloundou et al (2023**) https://openai.com/research/gpts-are-gpts
- **Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence by Erik Brynjolfsson et al (2025)** **h**ttps://digitaleconomy.stanford.edu/app/uploads/2025/11/CanariesintheCoalMine_Nov25.pdf
- **Experimental evidence on the productivity effects of generative artificial intelligence by** Noy, **Shakked ; Zhang, Whitney(2023)** https://www.science.org/doi/10.1126/science.adh2586; accessible via https://idiscover.lib.cam.ac.uk/permalink/f/1kas1sp/TN_cdi_crossref_primary_10_1126_science_adh2586

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